

1.

**In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.**

At time t seconds, $t \geq 0$, a particle P is moving with velocity $\mathbf{v} \text{ ms}^{-1}$, where

$$\mathbf{v} = 3(t+2)^2 \mathbf{i} + 5t(t+2) \mathbf{j}$$

Position vectors are given relative to the fixed point O

At time $t = 0$, P is at the point with position vector $(-30\mathbf{i} - 45\mathbf{j}) \text{ m}$.

(a) Find the position vector of P when $t = 3$

(4)

(b) Find the magnitude of the acceleration of P when $t = 3$

(4)

At time T seconds, P is moving in the direction of the vector $2\mathbf{i} + \mathbf{j}$

(c) Find the value of T

(2)

2. A particle Q of mass 3 kg is moving on a smooth horizontal surface.

Particle Q is moving with velocity $5\mathbf{i} \text{ ms}^{-1}$ when it receives a horizontal impulse of magnitude $3\sqrt{82} \text{ N s}$.

Immediately after receiving the impulse, the velocity of Q is $(x\mathbf{i} + y\mathbf{j}) \text{ ms}^{-1}$, where x and y are positive constants.

The kinetic energy **gained** by Q as a result of receiving the impulse is 138 J.

Find, in terms of $\mathbf{i}$ and $\mathbf{j}$, the velocity of Q immediately after receiving the impulse.

(7)

3.

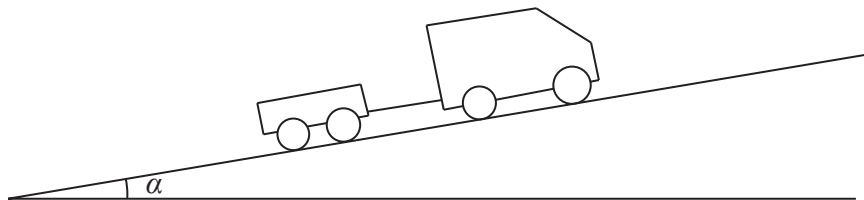


Figure 1

A van of mass 900 kg is moving up a straight road inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{25}$. The van is towing a trailer of mass 200 kg. The trailer is attached to the van by a rigid towbar which is parallel to the direction of motion of the van and the trailer, as shown in Figure 1.

The resistance to the motion of the van from non-gravitational forces is modelled as a constant force of magnitude 400 N.

The resistance to the motion of the trailer from non-gravitational forces is modelled as a constant force of magnitude 240 N.

The towbar is modelled as a light rod.

The engine of the van is working at a constant rate of 15 kW.

(a) Find the acceleration of the van at the instant when the speed of the van is 12 ms^{-1}

(5)

At the instant when the speed of the van is 14 ms^{-1} , the trailer is passing the point A on the slope and the towbar breaks.

The trailer continues to move up the slope until it comes to rest at the point B .

The resistance to the motion of the trailer from non-gravitational forces is still modelled as a constant force of magnitude 240 N.

(b) Use the work-energy principle to find the distance AB .

(4)

4.

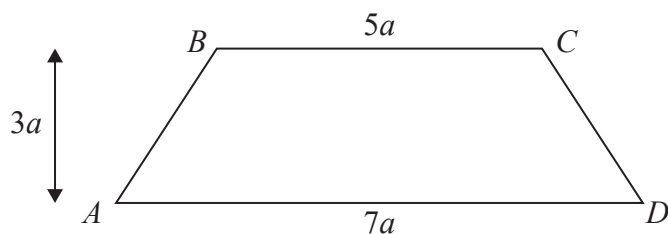


Figure 2

The uniform lamina $ABCD$ shown in Figure 2 is in the shape of an isosceles trapezium.

- BC is parallel to AD and angle BAD is equal to angle ADC
- $BC = 5a$ and $AD = 7a$
- the perpendicular distance between BC and AD is $3a$
- the distance of the centre of mass of $ABCD$ from AD is d

(a) Show that $d = \frac{17}{12}a$

(5)

The uniform lamina $PQRS$ is a rectangle with $PQ = 5a$ and $QR = 9a$.

The lamina $ABCD$ in Figure 2 is used to cut a hole in $PQRS$ to form the template shown shaded in Figure 3.

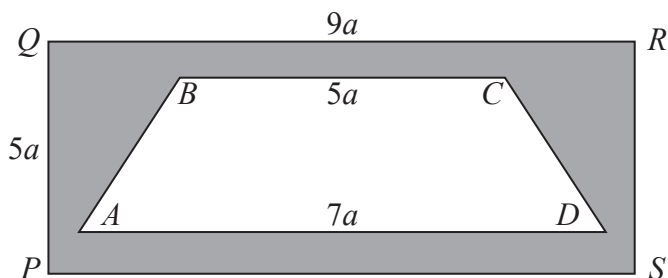


Figure 3

- PS is parallel to AD
- the perpendicular distance between PS and AD is a
- the perpendicular distance of A from PQ is a

The template is freely suspended from P and hangs in equilibrium with PS at an angle of θ° to the downward vertical.

(b) Find the value of θ

(7)

5. The fixed points X and Y lie on horizontal ground.

At time $t = 0$, a particle P is projected from X with speed $u \text{ ms}^{-1}$ at angle θ to the ground.

Particle P moves freely under gravity and first hits the ground at Y .

(a) Show that $XY = \frac{u^2 \sin 2\theta}{g}$ (6)

The points A and B lie on horizontal ground. A vertical pole CD has length 5 m. The end C is fixed to the ground between A and B , where $AC = 12 \text{ m}$.

At time $t = 0$, a particle Q is projected from A with speed 20 ms^{-1} at 60° to the ground.

Particle Q moves freely under gravity, passes over the pole and first hits the ground at B , as shown in Figure 4.

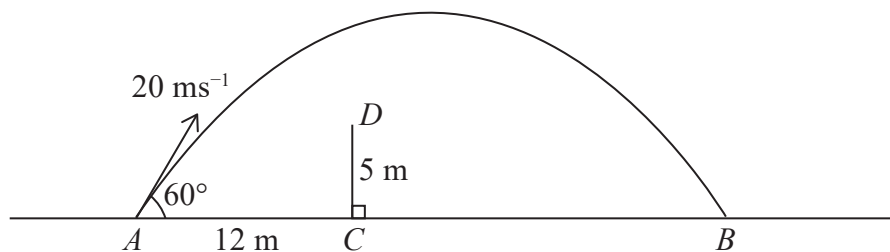


Figure 4

- (b) Find the distance CB . (2)
- (c) Find the height of Q above D at the instant when Q passes over the pole. (4)

6.

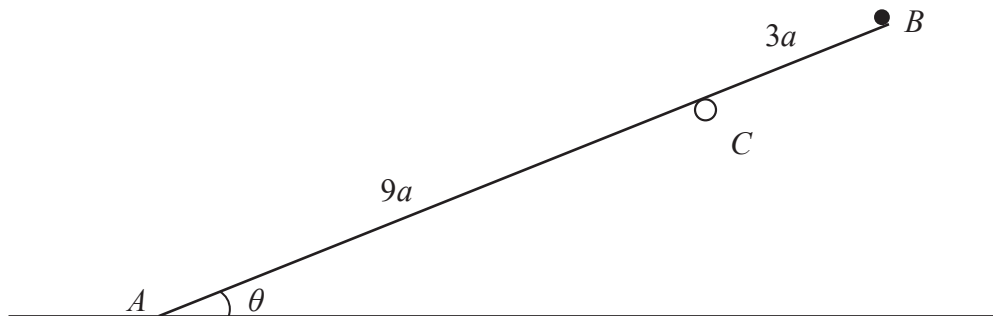


Figure 5

A uniform beam AB , of weight $5W$ and length $12a$, rests with end A on rough horizontal ground.

A package of weight W is attached to the beam at B .

The beam rests in equilibrium on a smooth horizontal peg at C , with $AC = 9a$, as shown in Figure 5.

The beam is inclined at an angle θ to the ground, where $\tan \theta = \frac{5}{12}$

The beam is modelled as a rod that lies in a vertical plane perpendicular to the peg.
The package is modelled as a particle.

The normal reaction between the beam and the peg at C has magnitude kW

Using the model,

(a) show that $k = \frac{56}{13}$ (4)

The coefficient of friction between A and the ground is μ

Given that the beam is resting in limiting equilibrium,

(b) find the value of μ (6)

7. A particle P has mass $5m$ and a particle Q has mass $2m$.

The particles are moving in opposite directions along the same straight line on a smooth horizontal surface.

Particle P collides directly with particle Q .

Immediately **before** the collision, the speed of P is $2u$ and the speed of Q is $3u$.

Immediately **after** the collision, the speed of P is x and the speed of Q is y .

The direction of motion of Q is **reversed** as a result of the collision.

The coefficient of restitution between P and Q is e .

- (a) Find the set of values of e for which the direction of motion of P is **unchanged** as a result of the collision.

(7)

In the collision, Q receives an impulse of magnitude $\frac{60}{7}mu$

- (b) Show that $e = \frac{1}{5}$

(5)

After the collision, Q hits a smooth fixed vertical wall that is perpendicular to the direction of motion of Q .

Particle Q rebounds and there is a second collision between P and Q .

The coefficient of restitution between Q and the wall is $\frac{1}{3}$

- (c) Find, in terms of m and u , the magnitude of the impulse received by Q in the second collision between P and Q .

(3)